

Sachin Anand

☎ +91 93545 80732

✉ sachina2020@iitkalumni.org | ssachin919@gmail.com

Research Profile

Research-oriented engineer with 4+ years of experience building AI-driven systems, data pipelines, and automation frameworks that accelerate experimentation and analysis. Strong background in applied machine learning, knowledge graphs, and AI agents for research workflows. Experienced in translating abstract ideas into working prototypes, datasets, and evaluation pipelines. IIT Kanpur alumnus seeking a Research Assistantship in applied AI, computational research, or data-driven social and engineering domains.

Research Interests

- Applied Artificial Intelligence and Machine Learning Systems
- Large Language Models, RAG, and Knowledge Graphs
- AI Agents for Research Automation and Experimentation
- Computational Social Science and Network Analysis
- Data-driven Modeling and System Prototyping

Education

- **Indian Institute of Technology Kanpur**
B.Tech, Mechanical Engineering 2016 – Dec 2020

Technical Skills

- **Programming:** Python, JavaScript, Java, C++
- **AI / ML:** LangGraph, LangChain, RAG, Prompt Engineering, Context Engineering
- **Data & Graphs:** PostgreSQL, MongoDB, Neo4j, Pinecone
- **Backend & APIs:** FastAPI, Django, Flask, Node.js
- **Cloud & Compute:** AWS, GCP, Azure, Serverless Architectures
- **DevOps & Automation:** Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Azure Pipelines)
- **Research Tooling:** Experiment pipelines, ETL workflows, dataset preparation

Research & Professional Experience

Webyalaya (Co-founder & Tech Lead) *Sep 2025 – Present*
Peer-to-Peer Learning Platform — Incubated by IIM Lucknow

- Co-founded a peer-to-peer learning platform incubated at **IIM Lucknow**, one of India's premier management institutions, validating the startup's potential and go-to-market strategy.
- Architected the full-stack infrastructure: Next.js/Vercel frontend, AWS ECS/Fargate backend, Azure PostgreSQL, LiveKit on Azure VMs for real-time sessions, AWS S3, and Cloudflare + GitHub Actions CI/CD pipeline.
- Grew the platform to **100+ active users** through word-of-mouth alone — zero paid marketing — demonstrating strong organic product-market fit.
- Built and managed a team of **10–12 interns** who made meaningful product contributions while gaining hands-on startup experience.
- Maintained a **50/50 gender ratio** in hiring from inception; the startup is **woman-led**, with a female co-founder driving key product and growth decisions.

TalentSavvy Software Private Limited
Research-Oriented Software Engineer (Intern)

May 2025 – Sep 2025

- Built data pipelines and ETL workflows in Python to support large-scale analytics and experimentation.
- Designed an AI-driven agent using LangGraph to automate build, test, and deployment experiments, enabling faster iteration cycles.
- Conducted prompt and context engineering to systematically improve LLM-based agent accuracy and reliability.
- Developed internal tooling to track experiment outcomes and system performance across multiple environments.

Independent Research Projects & Startup Labs

Aug 2022 – Apr 2025

Applied AI & Systems Researcher

- **AI-driven Education Platforms** — Designed adaptive learning systems using RAG and conversational interfaces; analyzed learner behavior data to improve engagement and completion rates.
- **Knowledge Graph Systems** — Built graph-backed retrieval systems (Neo4j, Pinecone) to structure unorganized data for semantic querying and downstream analysis.
- **Applied AI Prototypes (Hackathons)** — Rapidly prototyped AI-powered systems under constrained timelines, focusing on feasibility validation, system architecture, and evaluation.
- **Blockchain & Data Integrity Research** — Implemented decentralized verification mechanisms for trust, provenance, and fraud detection in experimental platforms.

Contata Solutions

Aug 2021 – Aug 2022

System Analyst

- Developed data-backed matching and recommendation logic using graph databases and relational models.
- Designed backend systems for civic-scale datasets, enabling structured reporting, tracking, and analysis of public issues.
- Automated deployment and testing workflows to ensure reproducibility and reliability of production systems.

Selected Coursework & Training

- Knowledge Graphs for RAG — DeepLearning.AI
- Deep Learning Specialization — Andrew Ng (Coursera)
- An Introduction to Artificial Intelligence — NPTEL (Prof. Mausam)
- Google Kubernetes Engine — Google Cloud
- Blockchain & Cryptocurrency Program — IIT Madras CCE